

UD65640

Application Note

BIST Test

AN-K0006-240805

1. Target	3
2. Register	3
ACCESS CONTROL: MANUFACTURER COMMAND ACCESS CONTROL (80H)	3
BIST_CTRL: (F0H)	3
BIST_CFG: (F1H).....	5
3. HW BIST Pin Config.....	6
4. HW BIST Power Sequence	7
5. Flow: Register Control BIST	8
6. Flow: HW BIST.....	9
7. Example: Register Control	10
8. Revision History	12

1. Target

UD65640 include build in self-test mode for test panel display, Function enable of use register setting or HW pin control

2. Register

Access Control: Manufacturer Command Access Control (80h)

80h	Access Control									
	W/R	B7	B6	B5	B4	B3	B2	B1	B0	Hex
Command	—	1	0	0	0	0	0	0	0	80h
Parameter 1	W/R	0	0	0	0	0	0	0	MCS_EN	00h
Descriptions	MCS_EN: This command releases parameters so that Manufacturer Command Access are enabled. 1: Access enabled. 0: Access denied. (Protection on) 1 = Must fill 1; 0 = Must fill 0; X = Don't care.									
Restrictions										

BIST_CTRL: (F0h)

F0h	BIST_CTRL									
	W/R	B7	B6	B5	B4	B3	B2	B1	B0	Hex
Command	—	1	1	1	1	0	0	0	0	F0h
Parameter 1	W/R	BIST_SEL[4:0]					0	0	BIST_EN	00h
Parameter 2	W/R	0					AGING_EN	AGING_PRD [1:0]		
Descriptions	BIST_EN: Build in self-test function, 1: enable / 0: disable.									
	BIST_SEL [4:0]: Patten Select									
	BIST Pattern								BIST_SEL[4:0](HEX)	
	1	L255						0x0		
	2	L0						0x1		
	3	Red						0x2		
	4	Green						0x3		

The information contained herein is confidential and an exclusive property of UltraDisplay Inc. and shall not be distributed, or disclosed in whole or in part without prior permission of UltraDisplay Inc.

	5	Blue		0x4
	14	Gray		0xD
	15	Horizontal line inversion		0xE
	16	Vertical line inversion		0xF
	17	Crosstalk – V1		0x10
	18	Crosstalk – V2		0x11
	19	Crosstalk – V3		0x12
	20	Crosstalk – H1		0x13
	21	Crosstalk – H2		0x14
	22	Crosstalk – H3		0x15
	23	Gradient + 8 Color Bar		0x16
	26	checkerboard		0x19
	27	Dot inversion		0x1A
	28	Boundary		0x1B
AGING_EN: AGING Mode function, 1: enable / 0: disable. Fix patten sequence: White > Black > Red > Green > Blue > V Gradient Gray > Crosstalk > Color bar. AGING_PRD [1:0]: AGING Mode patten change time. b'00: 31 frames b'01: 63 frames b'10: 95 frames b'11: 127 frames 1 = Must fill 1; 0 = Must fill 0; X = Don't care.				
Restrictions				

BIST_CFG: (F1h)

F1h	BIST CGF									
	W/R	B7	B6	B5	B4	B3	B2	B1	B0	Hex
Command	—	1	1	1	1	0	0	0	1	F1h
Parameter 1	W/R	BIST_PIXEL0_R[7:0]								FFh
Parameter 2	W/R	BIST_PIXEL0_G[7:0]								FFh
Parameter 3	W/R	BIST_PIXEL0_B[7:0]								FFh
Parameter 4	W/R	BIST_PIXEL1_R[7:0]								00h
Parameter 5	W/R	BIST_PIXEL1_G[7:0]								00h
Parameter 6	W/R	BIST_PIXEL1_B[7:0]								00h
Descriptions	BIST_PIXEL: BIST Patten Data Value, only active List Patten									
	BIST Pattern		BIST_SEL [4:0]	PIXEL0_R	PIXEL0_G	PIXEL0_B	PIXEL1_R	PIXEL1_G	PIXEL1_B	
	14	Gray		0xD	0 ~ 255	0 ~ 255	0 ~ 255	Not use	Not use	Not use
	15	Horizontal line inversion		0xE	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255
	16	Vertical line inversion		0xF	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255
	17	Crosstalk – V1		0x10	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255
	18	Crosstalk – V2		0x11	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255
	19	Crosstalk – V3		0x12	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255
	20	Crosstalk – H1		0x13	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255
	21	Crosstalk – H2		0x14	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255
	22	Crosstalk – H3		0x15	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255
	26	checkerboard		0x19	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255
	27	Dot inversion		0x1A	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255

The information contained herein is confidential and an exclusive property of UltraDisplay Inc. and shall not be distributed, or disclosed in whole or in part without prior permission of UltraDisplay Inc.

28	Boundary		0x1B	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255	0 ~ 255
1 = Must fill 1; 0 = Must fill 0; X = Don't care.									
Restrictions									

3. HW BIST Pin Config

HW BIST only input power and follow sequence can panel run BIST patten.

No.	Symbol	Description
50	EXT_PWR	Open
49	Reserved	Open
48	ELVDD	Power Supply 1.2V
47	ELVSS	Power Supply -6V
46	AVCI18	Power Supply 1.8V
45	Reserved	Open
44	Reserved	Open
43	Reserved	Open
42	RESX	Connect IOVCC
41	AVCI18	Power Supply 1.8V
40	DVDD	Open / Output1.2V
39	DVSS	GND
38	AVSS	GND
37	AVCI33	Power Supply 3.3V
36	Reserved	Open
35	Reserved	Open
34	Reserved	Open
33	Reserved	Open
32	Reserved	Open
31	Reserved	Open
30	Reserved	Open
29	Reserved	Open
28	Reserved	Open
27	Reserved	Open
26	Reserved	Open
25	Reserved	Open
24	Reserved	Open
23	Reserved	Open
22	Reserved	Open
21	BIST_EN	Connect IOVCC
20	BIST_SEL0	Connect IOVCC or Open
19	BIST_SEL1	Connect IOVCC or Open
18	BIST_SEL2	Connect IOVCC or Open
17	BIST_SEL3	Connect IOVCC or Open
16	Reserved	Open
15	BIST_SEL4	Connect IOVCC or Open
14	Reserved	Open
13	Reserved	Open
12	Reserved	Open
11	Reserved	Open
10	Reserved	Open
9	Reserved	Open

The information contained herein is confidential and an exclusive property of UltraDisplay Inc. and shall not be distributed, or disclosed in whole or in part without prior permission of UltraDisplay Inc.

8	Reserved	Open
7	Reserved	Open
6	Reserved	Open
5	IOVCC	Power Supply 3.3V
4	ELVSS	Power Supply -6V
3	ELVDD	Power Supply 1.2V
2	Reserved	
1	Reserved	

Table 1

4. HW BIST Power Sequence

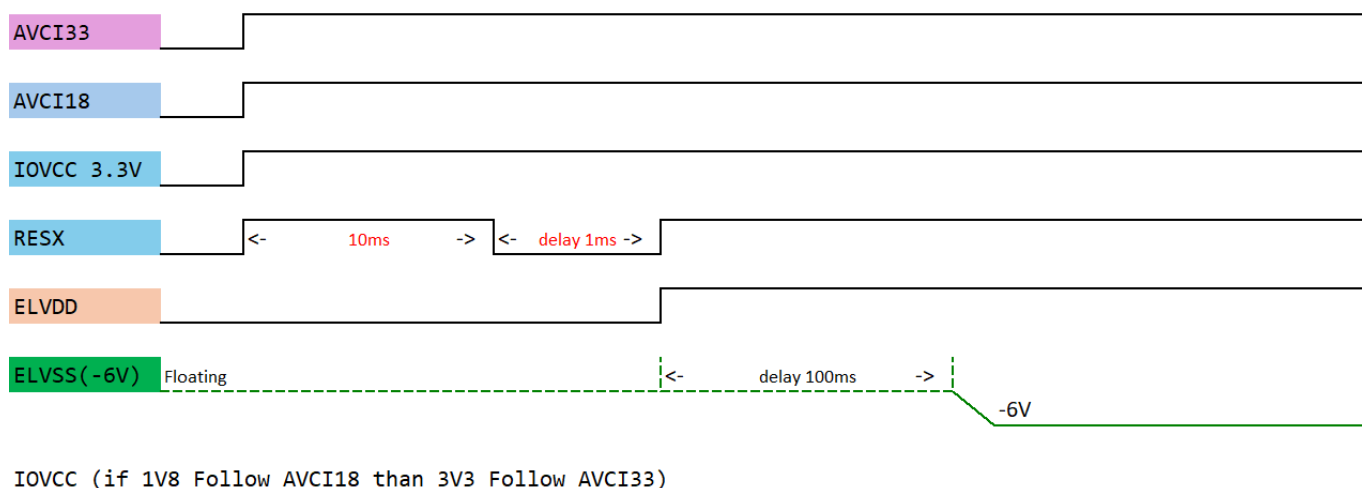


Figure 1

5. Flow: Register Control BIST

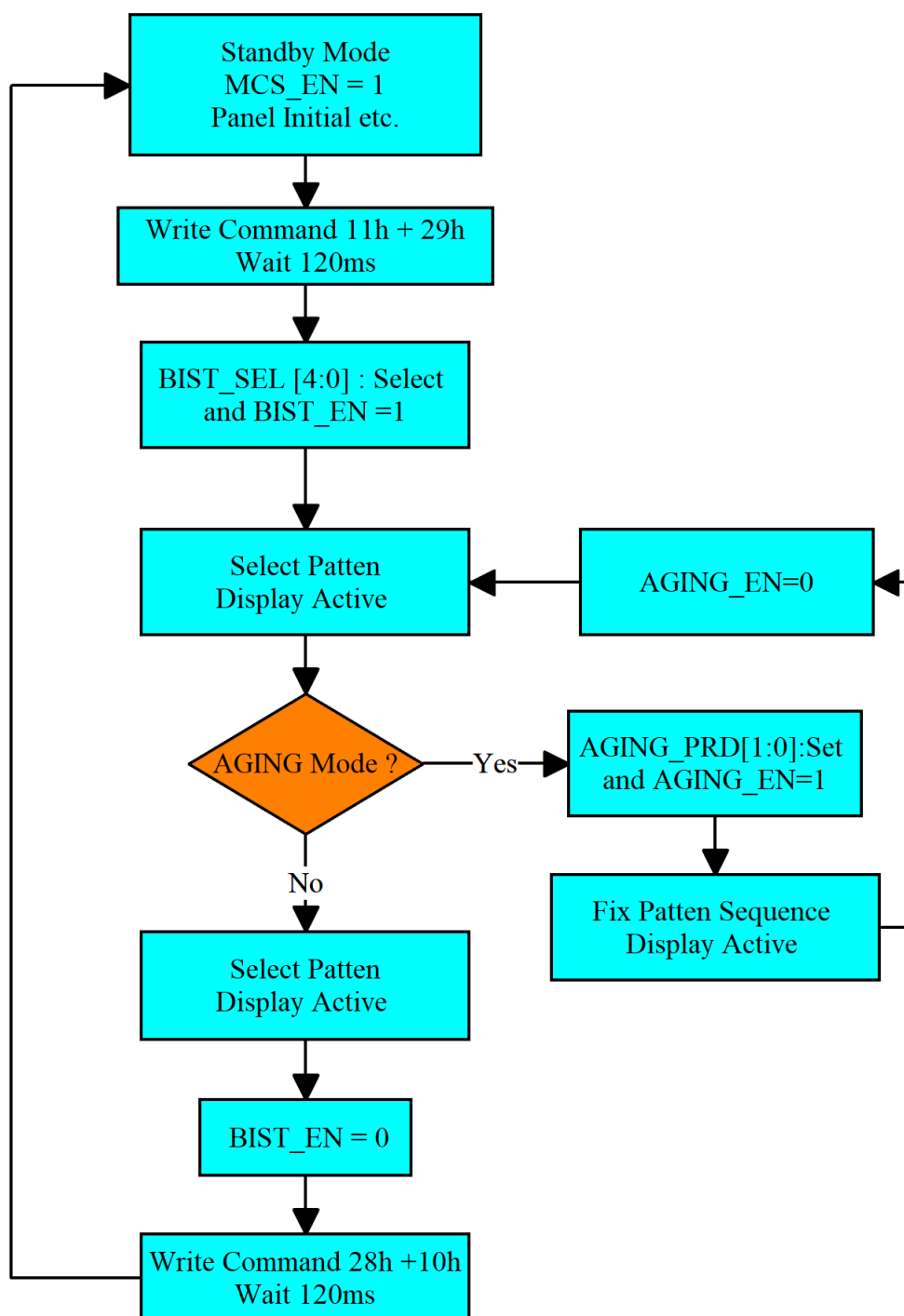


Figure 2

6. Flow: HW BIST

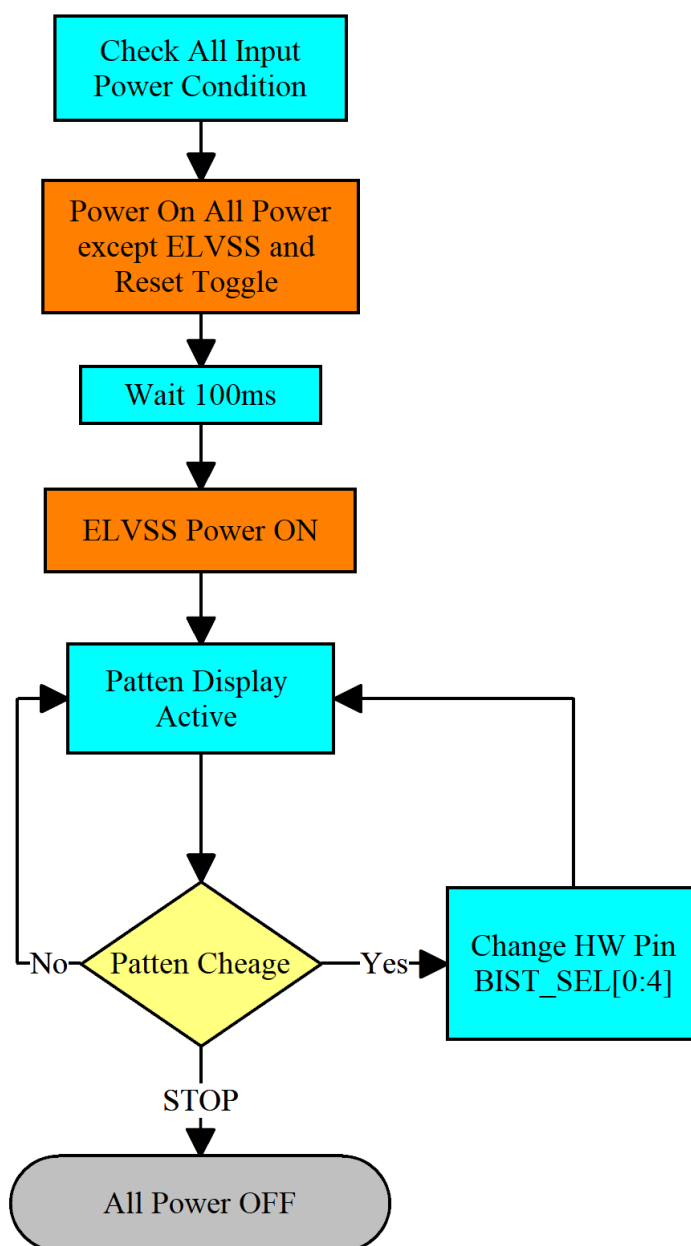


Figure 3

7. Example: Register Control

Step1. Power on and check all input voltage.

Step2. Initial panel setting(Only Command Mode), now status standby mode.

Step3. Write command 11h.

Step4. Write command 29h.

Step5. Wait 120ms.

Step6. Write Command 80h,01h

Step7. Write Command F0h, B1h // Gradient + 8 Color Bar BIST_SEL [4:0] = 0x16.

Step8. Panel display Gradient + 8 Color Bar patten, now status active mode.

Step9. Write Command F0h, 00h //BIST EN = 0

Step10. Write command 28h.

Step11. Write command 10h.

Step12. Wait 120ms.

Step13. now status standby mode.

8. Example: Gray Pattern Chage DATA

9. Step1. Power on and check all input voltage.
10. Step2. Initial panel setting (Only Command Mode), now status standby mode.
11. Step3. Write command 11h.
12. Step4. Write command 29h.
13. Step5. Wait 120ms.
14. Step6. Write Command 80h,01h
15. Step7. Write Command F1h, FFh, FFh, FFh, 00h, 00h, 00h
16. Step8. Write Command F0h, 69h //Gray BIST_SEL [4:0] = 0xD.
17. Step9. Panel display Gray (DATA=0xFF) patten, now status active mode.
18. Step10. Write Command F1h, EFh, EFh, EFh, 00h, 00h, 00h
19. Step11. Panel display Gray (DATA=0xEF) patten, now status active mode.
20. Step12. Write Command F1h, DFh, DFh, DFh, 00h, 00h, 00h
21. Step13. Panel display Gray (DATA=0xDF) patten, now status active mode.
22. Step14. Write Command F1h, CFh, CFh, CFh, 00h, 00h, 00h
23. Step15. Panel display Gray (DATA=0xCF) patten, now status active mode.
24. Step16. Write Command F0h, 00h //BIST EN = 0
25. Step17. Write command 28h.
26. Step18. Write command 10h.
27. Step19. Wait 120ms.
28. Step20. now status standby mode.

29. Revision History

Version No.	Date	Page	Description
AN-K0006-240805	2024/8/5	12	Initial Version